

Daily Epsilon

My solutions to the daily maths problems from your Daily Epsilon of Math calender.

Day 7

$$-\frac{4}{3}(x-25) = \frac{3}{4}(x+25)$$

Solution

Multiply both sides by 12;

$$12 \cdot -\frac{4}{3}(x-25) = 12 \cdot \frac{3}{4}(x+25)$$

$$4 \cdot -4(x-25) = 3 \cdot 3(x+25)$$

$$-16(x-25) = 9(x+25)$$

$$-16x + 400 = 9x + 225$$

$$-16x - 9x = 225 - 400$$

$$-25x = -175$$

$$x = 7$$

Day 9

Find the number of digits in;

$$14^{22} \cdot 7^{-20}$$

Solution



Figure 1: A screenshot of a toot of mine that i made on mastodon.

Of course, stupid me didn't read the instructions. I just saw the the figures and i dived strstight into solving. i got to a point and i wondered to myself that this doesn't make any sense at all. I then checked and saw that i was supposed to find the number of digits in the *equation*. A little simplification before jumping into finding the number of digits;

$$(7 \cdot 2)^{22} \cdot 7^{-20}$$

$$(7^{22} \cdot 2^{22}) \cdot 7^{-20}$$

$$\text{recall that : } a^b + a^c = a^{b+c}$$

$$\therefore (7^{20} \cdot 7^2 \cdot 2^{22}) \cdot 7^{-20}$$

$$\text{recall that : } \left(a^{-1} = \frac{1}{a}\right)$$

$$\therefore (7^{20} \cdot 7^2 \cdot 2^{22}) \cdot \left(\frac{1}{7^{20}}\right)$$

$$(\cancel{7^{20}} \cdot 7^2 \cdot 2^{22}) \cdot \left(\frac{1}{\cancel{7^{20}}}\right)$$

our simplification finally brings us here;

$$= 7^2 \cdot 2^{22}$$

Now, to find the length of the result, the only way i know how to do this is by writing some code that computes the powers of these integers and then prints the length of the result. My language of choice here is java. Yes, java. python, you say? nah.

Here goes the java code;

```

public class Epsilon {
    public static void main(String[] args) {
        int a = (int) Math.pow(7, 2);
        int b = (int) Math.pow(2, 22);

        int c = a * b;

        System.out.println(String.valueOf(c).length());
    }
}

```

I could inline the above code but it might make it difficult to look at than it already is if you are not familiar with the java programming language or programming in general.¹ When you run this in an Integrated Development Environment (IDE) that supports java, the result you get is 9.

$\therefore$ the length of the result of $7^2 \cdot 2^{22}$ is 9.

Day 14

$$\frac{95 + 87}{8 + 5}$$

Solution

i saw this problem and thought to myself, there had to be some catch that i am not aware of but there isn't actually. That's the question. Simplify, basically. So this is pretty straightforward, you have to sum the numbers in the numerator as well as the denominator.

$$\frac{95 + 87}{8 + 5} = \frac{182}{13} = 14$$

Day 16

find the number of integers that satisfy;

$$\frac{5}{y} \leq -0.3$$

Solution

$$\frac{y}{5} \leq -\frac{1}{0.3}$$

$$y \leq -\frac{5}{0.3}$$

¹inlining is basically cleaning your code where instead of having multiple lines, you can just have a single line like this;

```

public class Epsilon {
    public static void main(String[] args) {
        System.out.println(String.valueOf(Math.pow(7, 2) * Math.pow(2, 22)).length());
    }
}

```

$$y \leq -16.67$$

What is an integer? An integer is a whole number, meaning that it isn't a fraction. 0, -1, 2 are integers. 3.14, 2.713 aren't. From this we can already tell that -16.67 isn't the answer we are looking for but rather, the number of whole numbers that satisfy the equation.

Take **-16**: put it in the original equation and you have;

$$-0.31 \leq -0.3$$

which is essentially equal to the each other. Ergo, -16 satisfies the equation.

Take **-15**: repeat the same process from above and you have;

$$-0.33 \leq -0.3$$

which is true so -15 satisfies the equation.

Continuing to do this until you get to -1.

Take **-1**:

$$-5 \leq -0.3$$

which is also true. Take **0**:

$$0 \leq -0.3$$

This is false From -16 to -1, that is 16 integers so there are just 16 integers that satisfy the original equation.

Day 17

$$x > 0$$

$$\sqrt{21-x} + \sqrt{x+32} = 9$$

Solution

The first instinct here is to square both sides of course.

$$(\sqrt{21-x} + \sqrt{x+32})^2 = 9^2$$

Expanding the *R.H.S* of the above equation yields;²

$$(21-x) + 2\sqrt{(21-x)(x+32)} + (x+32) = 81$$

$$21 - x + x + 32 + 2\sqrt{(21-x)(x+32)} = 81$$

$$53 + 2\sqrt{(21-x)(x+32)} = 81$$

$$2\sqrt{(21-x)(x+32)} = 28$$

²I made a big mistake here when i attempted this problem on paper. I forgot that there were two duplicate terms that should be together i.e.

$$\sqrt{(21-x)(x+32)}$$

. It has been so long i did any serious math that i just made such serious error.

$$\sqrt{(21 - x)(x + 32)} = 14$$

squaring both sides;

$$(21 - x)(x + 32) = 196$$

Expanding the brackets;

$$672 - 11x - x^2 = 196$$

$$x^2 + 11x - 476 = 0$$

Using the Wolfram calculator;³

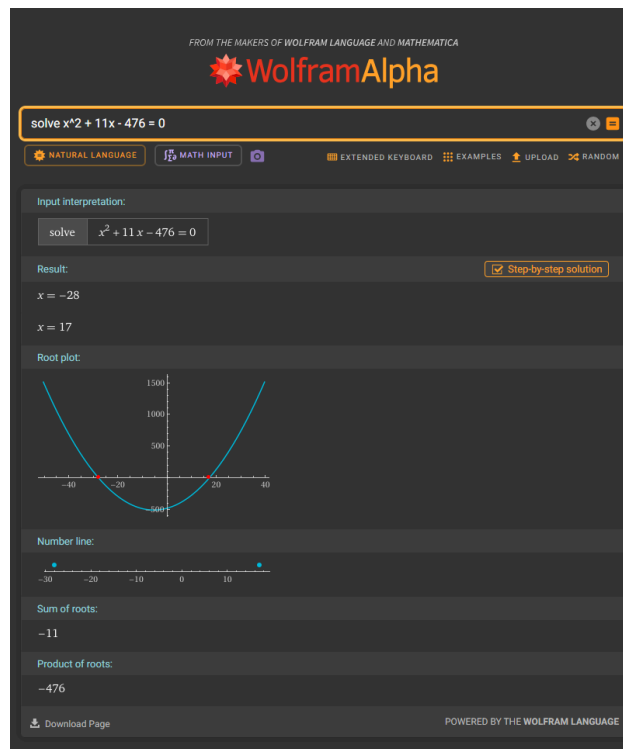


Figure 2: A screenshot of the results using Wolfram

Looking at Figure 2, we can see the results are $x = -28, 17$

since $x > 0$ therefore $x = 17$.

Day 20

The product of two rolls of a six-sided die is greater than 8 with a probability of $\frac{x}{36}$

Solution

Well, this is pretty straightforward, I believe. You have a single six-sided die and you roll it twice. With each roll of the dice, there are six possible outcomes so to find the possible products of the outcomes of the two rolls, we can make use of a table(6x6).

³At this point, I decided not to *waste* my time by doing the factorization looking for factors of 476 whose sums/ differences resulted in -11 so i pulled up Wolfram's calculator

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	8	10	12
3	3	6	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	6	12	18	24	30	36

Table 1: Table showing the products of the results of two rolls of a six-sided die.

Counting those products that are greater than 8. There are 20 of them. So $x = 20$

$\therefore$ the probability of the product of two rolls of a six-sided die being greater than 8 is $\frac{20}{36}$.

Day 22

$$+7 \cdot 10$$

$$-4 \cdot 8$$

$$+10 \cdot 4$$

$$-8 \cdot 7$$

Solution

Yup, you read that problem correctly. This was exactly the problem for the 22nd of March. This is pretty straightforward. So what we should do is bring out BODMAS⁴ and according to this rule, we should be handling all the multiplications before adding or subtracting. So doing this results in

$$= +70 - 32 + 40 - 56$$

$$= 110 - 88$$

$$= 22$$

⁴If this is your first time seeing this, allow me to explain the usefulness of Brackets, Of, Division, Multiplication, Addition, Subtraction. So

$$a + b \cdot c + d$$

gives you a different result if you add the terms together before multiplying them than if you multiplied terms together before then adding up. So to prevent from this variation in result, we have this in place. So you have to take the operations in this order. Why? I still do not know. I remember asking myself this question when i was teaching kids math during my service year. I should fix up on that.

Day 23

$$23^{29} \bmod 29$$

Solution

This is quite the problem. lol.⁵ I am currently struggling to solve this problem despite the approaches that i have taken.⁶

The first thought i had was writing a simple java program that found the value of 23 to the power of 29 using the function `pow()` imported from the `Math` class and then using the modulus operator to find the final answer but that didn't work because the data type i used for storing these values was 'int' and there is a maximum value that can be stored in int which is

$$2^{31}$$

and this is clearly smaller than 23 to the 29th power. I thought of using `BigInteger` but there is a problem in trying to use the modulus operator with the `BigInteger` variable.⁷

I tried using the modulus function for `BigInteger` but yet i still am not getting the answer which leaves me with the only other option of trying to breakdown the number into smaller bits that i can then perform computation on but 2 is clearly a prime which sucks. I might also try using the commutative property of modular arithmetic to break it down. I don't know. Lemme keep on trying. Several minutes later, and i have been rescued by the commutative power of modular arithmetic as well as a small java program that i wrote.

An important property of the remainder is that in addition, subtraction and multiplication, the remainder can be taken before the operation:

$$\begin{aligned}(a + b) \bmod m &= (a \bmod m + b \bmod m) \bmod m \\(a - b) \bmod m &= (a \bmod m - b \bmod m) \bmod m \\(a \cdot b) \bmod m &= (a \bmod m \cdot b \bmod m) \bmod m\end{aligned}$$

Thus, we can take the remainder after every operation and the numbers will never become too large.

Figure 3: A screenshot of an excerpt from the Competitive Programmer's Handbook written by Antti Laaksonen. (pp 6-7)

So just to refresh my knowledge of modular arithmetic, i consulted the Competitive Programmer's Handbook written by Antti Laaksonen and with that i just knew what to do. First, i had to break them down(ignoring the modulus for now);

$$\begin{aligned}23^{29} \\&= 23^{19} + 10 \\&= 23^{19} \cdot 23^{10} \\&= 23^9 \cdot 23^{10} \cdot 23^5 \cdot 23^5 \\&= 23^5 \cdot 23^4 \cdot 23^5 \cdot 23^5 \cdot 23^5 \cdot 23^5\end{aligned}$$

⁵Trying to also document my struggles with problems and the approaches that clearly didn't work.

⁶As i was writing this sentence, i suddenly had an a possible insight for an approach on how to go about solving the problem.

⁷If i haven't mentioned this or you are unfamiliar, there is an obvious clue to help you with solving the daily problems and that is the fact that the answer to the problem is that day's date. So going by this, The answer for today should be 23 and nothing else.

Now, bringing in the modulus part of the equation and using our updated knowledge on modular arithmetic from Figure 3.

$$= (23^5 \cdot 23^4 \cdot 23^5 \cdot 23^5 \cdot 23^5 \cdot 23^5) \bmod 29$$

$$= (23^5 \bmod 29 \cdot 23^4 \bmod 29 \cdot 23^5 \bmod 29 \cdot 23^5 \bmod 29 \cdot 23^5 \bmod 29 \cdot 23^5 \bmod 29) \bmod 29$$

We can now write a program that looks for the fifth and fourth power of 23 which would then find the modulus;

```
public class Day23 {

    public static void main(String[] args) {

        //finding the fourth and fifth power of 23 which then casts our answer to
        //type int because the type of the result of the pow() method is double.
        int h = (int) Math.pow(23, 5);
        int h1 = (int) Math.pow(23, 4);

        //finds the modulus of the results stored in the above variable.
        int i = h % 29;
        int i1 = h1 % 29;

        //multiplies the results.
        int c2 = (int) Math.pow(i, 5) * i1;

        //prints the result of c2 mod 29.
        System.out.println("the result of 23^29 mod 29 is " + c2%29);

    }
}
```

When you run this program, the printed statement on the terminal/console is “the result of 23^29 mod 29 is 23”. And there it is—our answer—23.

Day 24

A basketball team made 25% of its 3-point shots and 44% of its 2-point shots to score 40 points from 49 shots. How many 3-point shots were attempted?

Solution

After reading this question hundreds and a page filled with terrible attempts at a solution, i can finally say i have solved it and all i needed was a piss break. I took a piss break, came back and started afresh. Now to the solution.

49 shots were made consisting of 3 points shot and 2 point shots. Let

x = amount of 3pts shot attempted.

y = amount of 2pts shot attempted.

Given this, we can develop the following equation knowing the total shots made which is 49;

$$x + y = 49$$

Of the total amount of 3pts shots attempted, only 25% went in and of the total amount of 2pts shots attempted, only 44% went in. The total points scored from the 49 shots is 40 points which would mean that just 25% of the 3 points shots attempted summed together with 44% of the 2 points shots attempted result in a total score of 40. Given this, we can develop the following equation;

$$25\%(3x) + 44\%(2y) = 40$$

$$0.25(3x) + 0.44(2y) = 40$$

$$0.75x + 0.44y = 40$$

We now have a set of simultaneous equations.

$$x + y = 49 \rightarrow \text{eqn 1}$$

$$0.75x + 0.44y = 40 \rightarrow \text{eqn 2}$$

We will solve for just x since that is what we are concerned with. So making y the subject of formula in eqn 1 and substituting for y in eqn 2.

From eqn 1, we have $y = 49 - x$ and then substituting for y in eqn 2

$$0.75x + 0.88(49 - x) = 40$$

$$0.75x + 43.12 - 0.88x = 40$$

$$0.75x - 0.88x = 40 - 43.12$$

$$-0.13x = -3.12$$

$$x = \frac{3.12}{0.13}$$

$$x = 24$$

$\therefore$ The amount of 3-points shots attempted was 24.

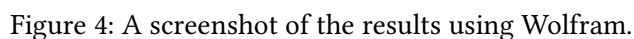
Day 26

$$\sum_{n=1}^6 \tan^2\left(\frac{n\pi}{14}\right)$$

Solution

$$\begin{aligned} \sum_{n=1}^6 \tan^2\left(\frac{n\pi}{14}\right) &= \tan^2\left(\frac{\pi}{14}\right) + \tan^2\left(\frac{2\pi}{14}\right) + \tan^2\left(\frac{3\pi}{14}\right) + \tan^2\left(\frac{4\pi}{14}\right) + \tan^2\left(\frac{5\pi}{14}\right) + \tan^2\left(\frac{6\pi}{14}\right) \\ &= \tan^2\left(\frac{\pi}{14}\right) + \tan^2\left(\frac{\pi}{7}\right) + \tan^2\left(\frac{3\pi}{14}\right) + \tan^2\left(\frac{2\pi}{7}\right) + \tan^2\left(\frac{5\pi}{14}\right) + \tan^2\left(\frac{3\pi}{7}\right) \end{aligned}$$

At this point, i used the calculator of my phone to compute the values and for the first three terms of the equation, i noticed that the results were too low that if i were to sum them all up, i wouldn't get

[illegible]
$$\therefore \sum_{n=1}^6 \tan^2\left(\frac{n\pi}{14}\right) = 26$$

$$\therefore \sum_{n=1}^6 \tan^2\left(\frac{n\pi}{14}\right) = 26$$

A blue trapezoid with a black outline. The top edge is shorter than the bottom edge, and the two side edges are slanted inwards from bottom to top.

⁸or maybe i didn't write the expression well in the calculator.

Day 31

Find the mode of the number of days in the months of 2026.

Solution

So this is a pretty interesting question. We could start by listing out the different months and the number of days in them or better still, list the days and then the different months with this number alongside their frequency. There are only three possible numbers 28, 30 and 31.

- 28 -> February(1).
- 30 -> April, June, September, November(4).
- 31 -> January, March, May, July, August, October, December(7).

∴ The mode of the number of days in the month of 2026 is 7.

April

Day 2

$$z + \frac{1}{z} = \sqrt{2}$$

$$u = z^{11} + z^{-11}$$

$$v = z^7 + z^{-7}$$

$$x = -(u \cdot v)$$

Solution

Let the first equation be equation 1, the second equation 2 and the third equation 3.
Squaring both sides of equation 1.

$$\left(z + \frac{1}{z}\right)^2 = (\sqrt{2})^2$$

$$z^2 + 2 + \frac{1}{z^2} = 2$$

$$z^2 + \cancel{2} + \frac{1}{z^2} = \cancel{2}$$

$$z^2 + \frac{1}{z^2} = 0 \tag{4}$$

The next thing to do now would be to square both sides of Equation 4.

$$\left(z^2 + \frac{1}{z^2}\right)^2 = 0$$

$$z^4 + 2 + \frac{1}{z^4} = 0$$

$$z^4 + \frac{1}{z^4} = -2 \tag{5}$$

Now, multiplying equation 2 by equation 1:

$$\left(z^2 + \frac{1}{z^2}\right)\left(z + \frac{1}{z}\right) = 0 \cdot \sqrt{2}$$

$$z^3 + z + \frac{1}{z} + \frac{1}{z^3} = 0$$

from Equation 1, we know that $\left[z + \frac{1}{z} = \sqrt{2}\right]$, so therefore

$$z^3 + \sqrt{2} + \frac{1}{z^3} = 0$$

$$z^3 + \frac{1}{z^3} = -\sqrt{2} \quad (6)$$

So now, finding the product of Equation 6 and Equation 5;

$$\left(z^4 + \frac{1}{z^4}\right)\left(z^3 + \frac{1}{z^3}\right) = -2 \cdot -\sqrt{2}$$

$$\left(z^4 + \frac{1}{z^4}\right)\left(z^3 + \frac{1}{z^3}\right) = 2\sqrt{2}$$

$$z^7 + z + \frac{1}{z} + \frac{1}{z^7} = 2\sqrt{2}$$

Again, from Equation 1, we know that $\left[z + \frac{1}{z} = \sqrt{2}\right]$, so therefore

$$z^7 + \sqrt{2} + \frac{1}{z^7} = 2\sqrt{2}$$

$$z^7 + \frac{1}{z^7} = 2\sqrt{2} - \sqrt{2}$$

$$z^7 + \frac{1}{z^7} = \sqrt{2} \quad (7)$$

$\therefore$ the value for v is $\sqrt{2}$

To find u, we need to find the product of Equation 7 and Equation 5;

$$\left(z^7 + \frac{1}{z^7}\right)\left(z^4 + \frac{1}{z^4}\right) = -2 \cdot \sqrt{2}$$

$$z^{11} + z^3 + \frac{1}{z^3} + \frac{1}{z^{11}} = -2\sqrt{2}$$

from Equation 6, we know that $\left[z^3 + \frac{1}{z^3} = -\sqrt{2}\right]$

$$z^{11} - \sqrt{2} + \frac{1}{z^{11}} = -2\sqrt{2}$$

$$z^{11} + \frac{1}{z^{11}} = -2\sqrt{2} + \sqrt{2}$$

$$z^{11} + \frac{1}{z^{11}} = -\sqrt{2}$$

$\therefore$ the value for u is $-\sqrt{2}$

$$x = -(u \cdot v)$$

$$x = -(-\sqrt{2} \cdot \sqrt{2})$$

$$x = \sqrt{2} \cdot \sqrt{2}$$

$$\therefore x = 2$$

Day 14

Find the greatest number of Mondays that can occur in three consecutive months.

Solution

So for this, you need the three months to have as much days in it that's possibly allowed. The maximum number of days you can have in a month is 31 and from the problem in Day 31 of March, you can observe that you can't have three consecutive months that have 31 days. It's either that there are separated by a month with 28/29 days or 30 days. So the next best option is to have two months with 31 days and a month with 30 days separating them. That's the first consideration for this problem. The next consideration for this problem is to make sure that the first day of the first month is on Monday.⁹

M	T	W	T	F	S	S
1						
8						
15						
22						
29	30	31				

Table 2: Month 1

M	T	W	T	F	S	S
			1	2	3	4
5						
12						
19						
26	27	28	29	30		

Table 3: Month 2

M	T	W	T	F	S	S
					1	2
3						
10						
17						
24						
31						

Table 4: Month 3

⁹The first day of the 29/30/31-day month usually ends up having 5 instances of that day so for example, if Monday is the first day of the month, there will be 5 Mondays in that month.

For the Tables 2, 3 and 4 which shows the three different months, we are only concerned with Mondays and then the days of the week which are dates ending the month and leading to the next month hence the sparse nature of the months.

Counting all of the Mondays in the above tables gives us a total of **14 Mondays which is the greatest number of Mondays that can occur in three consecutive months.**

Day 15

How many twin primes are under one hundred?

Solution

So what are twin primes? Twin primes are prime numbers that have a difference of just two. 2 and 3 can't be prime because their difference is just one. The twin primes under 100 are;

1. 3, 5.
2. 5, 7.
3. 11, 13.
4. 17, 19.
5. 29, 31.
6. 41, 43.
7. 59, 61.
8. 71, 73.

So in total, there are just **8** twin primes under 100.¹⁰

Day 16

$$18\sqrt{8} + 8\sqrt{18} = 15\sqrt{2x}$$

Solution

$$18\sqrt{4 \cdot 2} + 8\sqrt{9 \cdot 2} = 15\sqrt{2x}$$

$$18 \cdot 2\sqrt{2} + 8 \cdot 3\sqrt{2} = 15\sqrt{2x}$$

$$36\sqrt{2} + 24\sqrt{2} = 15\sqrt{2x}$$

$$60\sqrt{2} = 15\sqrt{2x}$$

$$\cancel{60}\sqrt{2} = \cancel{15}\sqrt{2x}$$

$$4\sqrt{2} = \sqrt{2x}$$

$$4\sqrt{2} = \sqrt{2} \cdot \sqrt{x}$$

$$\cancel{4}\cancel{\sqrt{2}} = \cancel{\sqrt{2}} \cdot \sqrt{x}$$

$$\sqrt{x} = 4^2$$

$$x = 16$$

¹⁰This is pretty weird. Haha. I had mentioned earlier on that the date is usually a clue as to what the answer for that day's problem would be. Well, not in this case. This was the exact opposite. Makes me wonder if what they meant was a thousand and not a hundred.

Future Joshua here: I checked and if you find the twin primes under 1000, it exceeds 15 so it couldn't have been a mistake. I guess my suspicion was wrong all along. Not all the problems have solutions that are exactly the date.

Day 20

A cup starts at 0°C and warms at 1°C per minute in a room that starts at 88°F and cools at 1°F per minute, both reading the same temperature after x mins.

Solution

Let the temperature that they both read after x mins be $z^{\circ}\text{C}$.

- Cup.
 - initial temperature, $t_i = 0^{\circ}\text{C}$
 - final temperature, $t_f = z^{\circ}\text{C}$
 - warming rate, $r_w = 1^{\circ}\text{C} / \text{min}$
- Room
 - initial temperature, $t_i = 88^{\circ}\text{F}$
 - final temperature, $t_f = z^{\circ}\text{C}$
 - cooling rate, $r_c = 1^{\circ}\text{F} / \text{min}$

First thing to do is to convert the initial temperature¹¹ of the room and the cooling rate from fahrenheit to celsius.

- Converting the initial temperature of the room.

Let the initial temperature of the room in celsius be y ,

$$\begin{aligned}\frac{y - 0}{100 - 0} &= \frac{88 - 32}{212 - 32} \\ \frac{y}{100} &= \frac{56}{180} \\ y &= \frac{56 \cdot 100}{180} \\ y &= 31\end{aligned}$$

$\therefore$ the initial temperature of the room in celsius is 31°C .

- Converting the cooling rate.

$$\begin{aligned}1^{\cancel{\circ}\text{F}} \cdot \frac{100^{\circ}\text{C}}{212^{\cancel{\circ}\text{F}}} \\ = 0.47 \frac{^{\circ}\text{C}}{\text{min}}\end{aligned}$$

$\therefore$ the cooling rate of the room in celsius is $31^{\circ}\text{C}/\text{min}$.

They both read the same temperature of $z^{\circ}\text{C}$ after x mins which would mean that z would be taken as a constant.

- For the cup,

¹¹The pendant in you might be thinking that since the room is cooling, shouldn't 88°F be the final temperature of the room and $z^{\circ}\text{C}$ be the initial temperature. I get it but *semantics*. Haha!

$$\begin{aligned}
r_w &= \frac{t_f - t_i}{t} \\
1 &= \frac{z - 0}{x} \\
1 &= \frac{z}{x} \\
z &= x
\end{aligned} \tag{1}$$

- For the room,

$$\begin{aligned}
r_c &= \frac{t_i - t_f}{t} \\
0.47 &= \frac{31 - z}{x} \\
0.47x &= 31 - z \\
z &= 31 - 0.47x
\end{aligned} \tag{2}$$

Equating Equation 1 and Equation 2;

$$\begin{aligned}
x &= 31 - 0.47x \\
x + 0.47x &= 31 \\
1.47x &= 31 \\
x &= \frac{31}{1.47} \\
x &= 21
\end{aligned}$$

∴ the cup and the room would both read the same temperature after 21 minutes.

Day 21

Find the smallest positive integer x such that $\frac{x}{672}$ is a terminating decimal.

Solution

First, we need to know what a terminating decimal is. A terminating decimal is a decimal that contains a finite number of digits after the decimal point. So we need to find the smallest positive integer x such that $\frac{x}{672}$ is a terminating decimal. For this, i have to bring out the big gun, JAVA. I have to write a program that finds me this number using a while loop.¹²

```
public class AprilDay21 {

    public static void main(String[] args) {
        int i = 1;

        float x = (float) 1 / 3; //fuck floats.
        double y = (double) 1 / 3; //yay doubles. lol
        double y1 = (double) 4 / 672;
        //it has been a while since i used floats and doubles so i have forgotten how
        much more accurate one is over the other.
        System.out.println(x);
    }
}
```

¹²The source code for the programs in this book can be found on this [Github repo](#).

```

    System.out.println(y);
    System.out.println(y1);
    //trying to see what the length is to determine at what point the decimal
    terminates in order to solve this question properly.
    //The length is 18 meaning y has 16 decimal places so i will be using this in
    the program i will write.
    System.out.println(String.valueOf(y).length());
    //the previous comment about the length being 18 isn't true because y1's
    length is 20
    System.out.println(String.valueOf(y1).length());

    int length = 18;

    while (length >= 18) {
        double z = (double) i / 672;
        length = String.valueOf(z).length();
        if (length >= 18) {
            i++;
        } else {
            System.out.println("the smallest possible integer, x, such that x/672
            is a terminating decimal is "+i);
        }
    }
}

```

∴ The smallest possible integer x such that $\frac{x}{672}$ is a terminating decimal is 21. That's the number that's you get when you run the above program.

Day 22

Find the number of digits(excluding trailing zeroes) to the right of the decimal point in

$$(0.8)^{20} \cdot (0.375)^{20}$$

Solution

Before diving into trying to find the answer, we need to first understand what trailing zeroes are. Trailing zeroes are sort of 0s that appear after a significant digit in a decimal number. Take for example, a number like 0.003453000, the three zeroes after the second three counting from left to right are considered trailing zeroes. So in this problem, if we happen to 11 zeroes after a significant digit, we won't consider that in the final answer which would be the number of digits to the right of the decimal point in the above *equation*.

Recall that;

$$a^b \cdot c^b = (a \cdot c)^b$$

So considering this, we can rewrite

$$(0.8)^{20} \cdot (0.375)^{20}$$

as

$$(0.8 \cdot 0.375)^{20}$$

Now, evaluating the last expression;

$$\begin{aligned}
 & 0.3^{20} \\
 &= 3.486784401 \cdot 10^{-11} \\
 &= 0.00000000003486784401
 \end{aligned}$$

There are no trailing zeroes so we can move on to count the number of digits to the right of the decimal point which after counting happens to be 20. $\therefore$ The number of digits(excluding trailing zeroes) to the right of the decimal point in $(0.8)^{20} \cdot (0.375)^{20}$ is 20.¹³

Day 25

¹³I actually am not losing my mind. I had gotten my answer but didn't feel too sure about it but i proceeded to write my solution despite my uncertainty which was also helped by the fact that the problem for April Day 20 had an answer of 21 and i know my solution was sound and logical. I went on mastodon only to then find in the replies of today's problem similar concerns of mine. I am not mad after all. Figure 5 shows the replies.

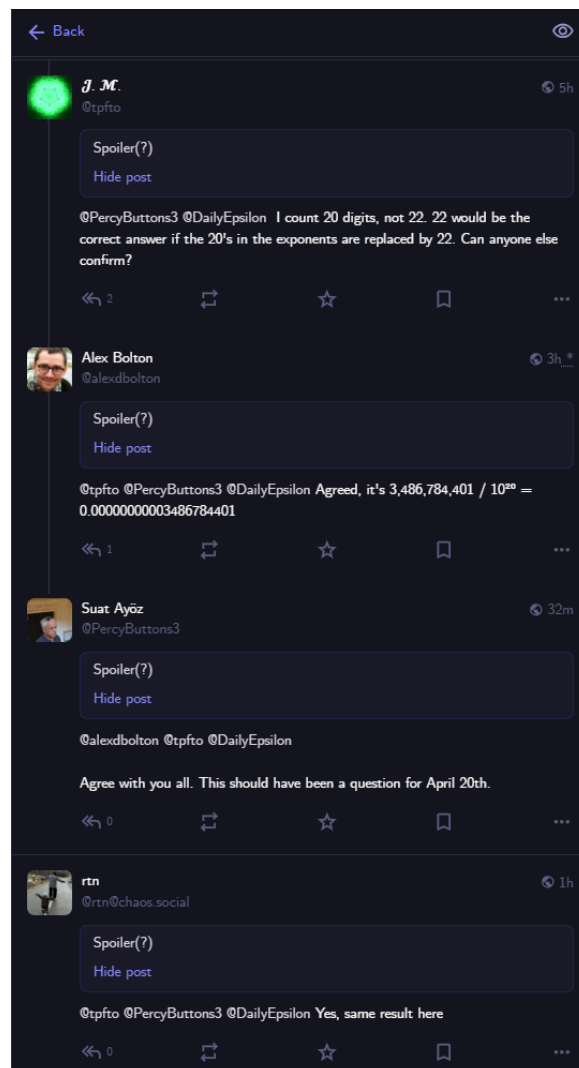


Figure 5: A screenshot of some of the replies to the problem of the day.

$$y = \sum_{i=1}^4 \frac{2i+1}{i^2(i+1)^2}$$

$$x = \frac{1}{1-y}$$

Solution

$$\sum_{i=1}^4 \frac{2i+1}{i^2(i+1)^2}$$

To get y , what we need to do is substitute one for i in the equation and do that for 2, 3 and 4 and then we sum everything up.

$$\begin{aligned} \sum_{i=1}^4 \frac{2i+1}{i^2(i+1)^2} &= \frac{2(1)+1}{(1)^2((1)+1)^2} + \frac{2(2)+1}{(2)^2((2)+1)^2} + \frac{2(3)+1}{(3)^2((3)+1)^2} + \frac{2(4)+1}{(4)^2((4)+1)^2} \\ &= \frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \frac{9}{400} \\ &= 0.96 \\ \therefore y &= 0.96 \end{aligned}$$

Substituting 0.96 for y in the second equation.

$$\begin{aligned} x &= \frac{1}{1-0.96} \\ x &= \frac{1}{0.04} \\ x &= 25 \end{aligned}$$

Day 26

45% of y is 18.

x is 65% of y .

Solution

45% of y being 18 gives us the following equation;

$$\begin{aligned} \frac{45}{100} \cdot y &= 18 \\ 45y &= 1800 \end{aligned} \tag{1}$$

x being 65% of y gives us the following equation;

$$\begin{aligned} y \cdot \frac{65}{100} &= x \\ y &= \frac{100x}{65} \end{aligned} \tag{2}$$

Substituting $\frac{100x}{65}$ for y in Equation 1.

$$\frac{45}{100} \cdot \frac{100x}{65} = 18$$

$$45x = 65 \cdot 18$$

$$45x = 1170$$

$$x = \frac{1170}{45}$$

$$x = 26$$

Day 27

Day 28

Find the smallest sum of three positive integers whose product is 792.

Solution

This shouldn't be too hard. What I could do is divide 792 by 2 which gives me 396 and keep on dividing. So basically trying to find the prime factors of 792.

$$\frac{792}{2} = 396$$

$$\frac{396}{2} = 198$$

$$\frac{198}{2} = 99$$

$$\frac{99}{3} = 33$$

$$\frac{33}{3} = 11$$

You can't divide further than this because obviously, 11 is a prime number so we can stop here.

$$\therefore 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 11 = 792$$

$$\text{also can be written as } 8 \cdot 9 \cdot 11 = 792$$

Summing all three numbers gives us a value of 28.

$$\text{also can be written as } 6 \cdot 11 \cdot 12 = 792$$

Summing all three numbers gives us a value of 29.

$$\text{also can be written as } 4 \cdot 11 \cdot 18 = 792$$

Summing all three numbers gives us a value of 33.

$$\text{also can be written as } 4 \cdot 6 \cdot 33 = 792$$

Summing all three numbers gives us a value of 43.

So far, the least sum of three positive integers whose product is 792 is 28.

∴ the smallest sum of three positive integers whose product is 792 is 28.

Day 29

$$\sum_{n=0}^{840} \frac{1}{\sqrt{n} + \sqrt{n+1}}$$

Solution

Unfortunately, i do not have any math tricks up my sleeve for this one and i can't compute the sum for this on paper so i will wield what i know well which is programming. Below is a simple java program that computes this.

```
public class April29 {
    public static void main(String[] args) {

        double total = 0;
        for(int n=0;n<=840;n++){
            double sum = (1/(Math.sqrt(n) + Math.sqrt(n+1)));
            total += sum;
        }
        System.out.println((int)total);
    }
}
```

When you run this program, you get 29 which is our answer.

Day 30

Find the number of four digit palindromes that are divisible by 3.

Solution

This isn't a hard problem. Time to bring the big man out, talking about java. I will write a simple program that will start with 1001 because that is the first four digit palindrome, and then loop(until we hit a 5 digit number which is 10000) over some lines of code which will reverse the number and store it in a different variable and then check with an if statement using the condition of if the number is equal to its reverse and if it is also divisible by 3. If this statement is true, a counter will be incremented by 1 and whether or not the condition is met, we will increment the number by 1.

```
public class AprilDay30 {

    public static void main(String[] args) {
        int number = 1001;
        int counter = 0;

        while (number != 10000) {
            String x = String.valueOf(number);
            StringBuilder y = new StringBuilder();

            for (int i = x.length() - 1; i > -1; i--) {
                y.append(x.charAt(i));
            }
        }
    }
}
```

```

        int convertedY = Integer.parseInt(String.valueOf(y));

        if ((number == convertedY) && (number % 3 == 0)) {
            counter++;
        }
        number++;
    }

    System.out.println("the total number of four digit palindromes divisible by 3
is " + counter);
}
}

```

After running this program, we'll get an output in th terminal which says that

the total number of four digit palindromes divisible by 3 is 30.

We can go a step further and list out all 30 numbers just for fun. If you include an ArrayList in our program and add all the valid numbers that meet the if condition, you can then print the list of these 30 numbers which are;

[1221, 1551, 1881, 2112, 2442, 2772, 3003, 3333, 3663, 3993, 4224, 4554, 4884, 5115, 5445, 5775, 6006, 6336, 6666, 6996, 7227, 7557, 7887, 8118, 8448, 8778, 9009, 9339, 9669, 9999]

May

Day 5

Find the number of values of

$$1 \leq n \leq 100$$

for which base-3 representations of both n and n^2 do not have the digit 2.

Solution

This is quite an interesting problem. It is now interesting how i see certain problems and just immediately think to myself that i can use code to solve this because this happens to be one of those problems. This is the case with this class of problems rather than the default which is to look for some way using the mathematical knowledge i have(which isn't a lot, mind you) which was basically my instinct before learning how to code.

I will be writing code for this. The pseudocode¹⁴ is presented below;

1. go into a for loop starting with the first number which is 1 up until 100
2. take number, n and square it, n^2
3. convert n and n^2 to base 3 and store in variables x and y.
4. using a function which checks if a particular digit is present in a string of digits;
 - check if 2 is present in x and store in a boolean variable b1 and,
 - check if 2 is present in y and store in a boolean variable b2 and,
5. if b1 or b2 is true, do nothing, else increment counter by 1.
6. print counter(after the loop ends).

My java¹⁵ code;

```
public class May5 {

    //this method checks for if there is a specified number is present in a long string of digits.
    public static boolean isNumberPresent(int number, char digit){
        String j = String.valueOf(number);
        boolean isPresent = false;

        for(int k=0;k<j.length();k++){
            if (j.charAt(k) == digit){
                isPresent = true;
                break;
            }
        }
        return isPresent;
    }

    public static void main(String[] args) {

        int counter = 0;

        for(int i=1;i<=100;i++){
            int n2 = i * i; //"i" is n while "n2" is n squared.

            int x = Integer.parseInt(Integer.toString(Integer.parseInt(String.valueOf(i), 10), 3));
```

¹⁴forgive my pseudocode. I plead for forgiveness in advance.

¹⁵I might include a javascript implementation of this algorithm later.

```

int y = Integer.parseInt(Integer.toString(Integer.parseInt(String.valueOf(n2), 10), 3));

boolean b1 = isNumberPresent(x, '2');
boolean b2 = isNumberPresent(y, '2');

if (b1 || b2){
    System.out.println(".....");
} else {
    counter++;
}
}

System.out.println(counter);
}
}

```

When you run this program, you get an output of 5. $\therefore$ the number of values of $1 \leq n \leq 100$ for which base-3 representations of both n and n^2 do not have the digit 2 is 5.

Day 14

$$\tan \theta + \cot \theta = 4$$

$$x = \tan^2 \theta + \cot^2 \theta$$

Solution

The first step to take would be squaring both sides of the first equation.

$$\tan \theta + \cot \theta = 4$$

$$(\tan \theta + \cot \theta)^2 = 4^2$$

$$(\tan \theta + \cot \theta)(\tan \theta + \cot \theta) = 4^2$$

$$\tan^2 \theta + \tan \theta \cot \theta + \tan \theta \cot \theta + \cot^2 \theta = 16$$

$$\tan^2 \theta + 2 \tan \theta \cot \theta + \cot^2 \theta = 16$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\therefore \tan^2 \theta + 2 \left(\frac{1}{\cot \theta} \right) \cot \theta + \cot^2 \theta = 16$$

$$\tan^2 \theta + 2 + \cot^2 \theta = 16$$

$$\tan^2 \theta + \cot^2 \theta = 16 - 2$$

$$\tan^2 \theta + \cot^2 \theta = 14$$

From the second equation in the question, we can see it is the exact same as the preceeding identity so $\therefore x = 14$.

Day 15

Two six-sided die rolls sum to a prime number with a probability $\frac{x}{36}$.

Solution

This is pretty straightforward. There is a similar problem that i solved from Day 20 back in March. We can make use of a similar approach from that solution here by drawing out a table.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Table 5: Table showing the sum of the results of two rolls of a six-sided die.

Counting the **bolded italic** numbers in the table which are obviously the prime numbers, we get a count of 15.

$\therefore$ Two six-sided die rolls sum to a prime number with a probability of $\frac{15}{36}$ which can also be reduced to $\frac{5}{12}$.

Day 20

If Jamie can bike three-quarters of a mile in 135 seconds, find how many miles she would travel in an hour.

Solution

This is a simple *physics* problem. You basically have to find the speed of Jamie and then using that, you can find out how much distance she would have covered in an hour.

$$s = \frac{d}{t}$$

where s is the speed, d is the distance and t is the time. using the parameters from the question, we can calculate the speed as follows;

$$s = \frac{0.75}{135} \text{ miles/s}$$

Now, we can calculate the distance as follows given that one hour in seconds is 3600s;

$$d = s \cdot t$$

$$d = \frac{0.75}{135} \cdot 3600$$

$$d = 20 \text{ miles}$$

$\therefore$ Jamie would cover 20 miles if she travels for one hour at constant speed.

Day 26

$$\frac{4 + \frac{1}{3}}{\frac{1}{2} - \frac{1}{3}}$$

Solution

Working first on the numerator and then the denominator.

$$\begin{aligned} 4 + \frac{1}{3} \\ \frac{12 + 1}{3} \\ = \frac{13}{3} \end{aligned}$$

The denominator;

$$\begin{aligned} \frac{1}{2} - \frac{1}{3} \\ \frac{3 - 2}{6} \\ = \frac{1}{6} \end{aligned}$$

Bringing the results gotten from the numerator and denominator.

$$\begin{aligned} \frac{13}{3} \div \frac{1}{6} \\ \frac{13}{3} \cdot \frac{6}{1} \\ \frac{13}{\cancel{3}} \cdot \frac{\cancel{6}}{1} \\ = 13 \cdot 2 \\ = 26. \end{aligned}$$

Day 29

$$38 + x + 71 + 12 + 62 + 88 = 300$$

Solution

This is the kind of problem you solve with your eyes closed. Haha. Even though, i do so, i still have to write out the solution.

$$38 + x + 71 + 12 + 62 + 88 = 300$$

adding everything on the left hand side of the equation and then solving for x ;

$$\begin{aligned} x + 271 &= 300 \\ x &= 300 - 271 \\ x &= 29 \end{aligned}$$

Day 31

x = the difference between the sum of the even and odd number divisors of 50.

Solution

First, we have to find divisors of 50 and separate them into the even number and odd number divisor sets.

The divisors are 1, 2, 5, 10, 25, 50.

The even number divisor set;

$$e = \{2, 10, 50\}$$

The sum of this set = $2 + 10 + 50 = 62$.

The odd number divisor set;

$$o = \{1, 5, 25\}$$

The sum of this set = $1 + 5 + 25 = 31$.

So, now finding the difference, d , of these sums;

$$\begin{aligned}d &= 62 - 31 \\d &= 31\end{aligned}$$

June

Day 4

$$4 \times 0.5 \times 8 \times 2 \times 0.125$$

Solution

This is pretty straightforward, I guess. We just multiply throughout but let's make this fun by replacing 0.5 with $\frac{1}{2}$ and 0.125 with $\frac{1}{8}$.

$$4 \times \frac{1}{2} \times 8 \times 2 \times \frac{1}{8}$$

$$4 \times \frac{1}{\cancel{2}} \times \cancel{8} \times \cancel{2} \times \frac{1}{\cancel{8}}$$

$$= 4$$

Day 8

$$\sqrt{100} + \sqrt{576} - \sqrt{100 + 576}$$

Solution

$$\sqrt{100} + \sqrt{576} - \sqrt{100 + 576}$$

$$10 + 24 - \sqrt{676}$$

$$34 - 26$$

$$= 8$$

Day 10

$$\sqrt{42 + 10\sqrt{17}} + \sqrt{42 - 10\sqrt{17}}$$

Solution

let's say

$$x = \sqrt{42 + 10\sqrt{17}} + \sqrt{42 - 10\sqrt{17}}$$

Squaring both sides of the equation

$$x^2 = \left(\sqrt{42 + 10\sqrt{17}} + \sqrt{42 - 10\sqrt{17}} \right)^2$$

$$\begin{aligned}
&\text{let } a = 42, b = 10\sqrt{17} \\
&\therefore x^2 = (\sqrt{a+b} + \sqrt{a-b})^2 \\
&\text{let } m = \sqrt{a+b}, n = \sqrt{a-b} \\
&\therefore x^2 = (m+n)^2 \\
&x^2 = m^2 + 2mn + n^2 \\
&\text{Replacing } m \text{ and } n \text{ with their respective values;} \\
&x^2 = (\sqrt{a+b})^2 + 2(\sqrt{a+b})(\sqrt{a-b}) + (\sqrt{a-b})^2 \\
&x^2 = a + b + 2\sqrt{(a+b)(a-b)} + a - b \\
&x^2 = a + b + 2\sqrt{(a^2 - b^2)} + a - b \\
&x^2 = 2a + 2\sqrt{(a^2 - b^2)} \\
&\text{Replacing } a \text{ and } b \text{ with their respective values;} \\
&x^2 = 2(42) + 2\sqrt{42^2 - (10\sqrt{17})^2} \\
&x^2 = 84 + 2\sqrt{1764 - (10 \times 17)} \\
&x^2 = 84 + 2\sqrt{1764 - 1700} \\
&x^2 = 84 + 2\sqrt{64} \\
&x^2 = 84 + 2 \times 8 \\
&x^2 = 84 + 16 \\
&x^2 = 100 \\
&x = \sqrt{100} \\
&x = 10
\end{aligned}$$

Day 15

Find the sum of all possible digits A such that the number 23A8 is divisible by 8.

Solution

We could manually do this with a calculator and start with 0 replacing A and then work our way through to 9 while keeping a set of all digits that let 23A8 be divisible by 6 but are we going to get what we want? Yes but is it elegant/beautiful? Not really. Why not write a program for this.

```

public class June15 {

    public static void main(String[] args) {
        String number = "23A8";

        int sum = 0;

        for (int i = 0; i <= 9; i++) {
            //You have to turn i into a string so that A in number can be replaced to perform the
            calculation.
            String y = String.valueOf(i);
            //Replacing A with the current number , i, in the loop.
            String newNumb = number.replace('A', y.charAt(0));
            //Turning newNumb to an integer because it is currently a string.
            int no = Integer.parseInt(newNumb);

```

```

        //checking if newNumb which is now stored in no, is divisible by 6 and if the condition is
true, we add it to the sum variable.
        if (no % 6 == 0) {
            sum = sum + i;
        }
    }

    System.out.println(
        "the sum of all possible digits A such that the number 23A8 is divisible by 6 is " + sum
    );
}
}

```

When you run this program, you get a sum of 15 with the following printed;

the sum of all possible digits A such that the number 23A8 is divisible by 6 is 15

Day 17

$$r + s + t = 5$$

$$\frac{1}{r} + \frac{1}{s} + \frac{1}{t} = 4$$

$$f(u, v) = \frac{u}{v} + \frac{v}{u}$$

$$x = f(r, s) + f(s, t) + f(t, r)$$

Solution

Given the above set of equations, we can say the following;

$$f(r, s) = \frac{r}{s} + \frac{s}{r}$$

$$f(s, t) = \frac{s}{t} + \frac{t}{s}$$

$$f(t, r) = \frac{t}{r} + \frac{r}{t}$$

Substituting these set of equations into x ;

$$\begin{aligned}
 x &= \frac{r}{s} + \frac{s}{r} + \frac{s}{t} + \frac{t}{s} + \frac{t}{r} + \frac{r}{t} \\
 x &= \frac{r^2t + s^2t + s^2r + t^2r + st^2 + sr^2}{srt} \\
 x &= \frac{s^2t + s^2r + r^2t + r^2s + t^2r + t^2s}{srt} \\
 x &= \frac{s^2(t + r) + r^2(t + s) + t^2(r + s)}{srt}
 \end{aligned}$$

going back to the very first equation ($r + s + t = 5$);

$$t + r = 5 - s$$

$$t + s = 5 - r$$

$$r + s = 5 - t$$

substituting these terms into x ;

$$x = \frac{s^2(5-s) + r^2(5-r) + t^2(5-t)}{srt}$$

$$x = \frac{5s^2 - s^3 + 5r^2 - r^3 + 5t^2 - t^3}{srt}$$

$$x = \frac{5(r^2 + s^2 + t^2) - (r^3 + s^3 + t^3)}{srt}$$

given that $r + s + t = 5$, we can find the square and cube of this equation;

Day 20

Find the number of even factors of 1620.

Solution

Here, we are looking for the total number of factors of 1620 that are even.¹⁶ So we can bring out the big gun—java—and write a simple program that consists of a *for* loop that ends when it gets to 1620 with an if condition within it that checks if the current number through the loop is divisible by 2 and if 1620 mod currentNumber is equal to zero. If the condition is met, the total variable increments by 1 and at the end of the program, the total is printed.

```
public class June20 {
    public static void main(String[] args) {
        int input = 1620;
        int total = 0;

        for (int l = 1; l <= input; l++) {
            if (l % 2 == 0 && input % l == 0) {
                total++;
            }
        }
        System.out.println(total);
    }
}
```

When you run the program, you get a total of 20.

∴ the number of even factors of 1620 is 20.

Day 21

¹⁶if you are wondering what this means, it is simple. A number that is divisible by 2.

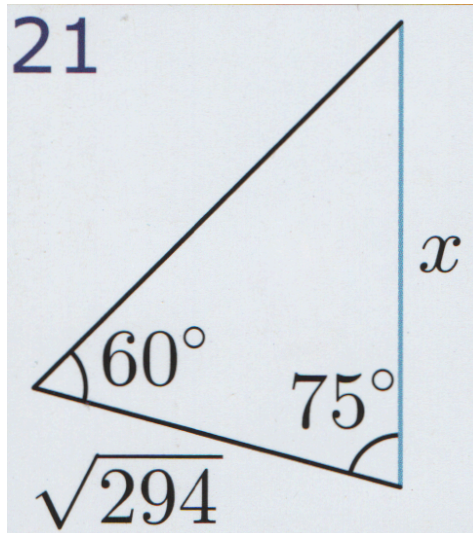


Figure 6: June 21 problem of the day¹⁷

Solution

To find x , we'll need to find the angle, θ , opposite the side of the triangle of length $\sqrt{294}$ units. The sum of the angles in a triangle is 180° so therefore;

$$\begin{aligned} 60^\circ + 75^\circ + \theta &= 180^\circ \\ 135^\circ + \theta &= 180^\circ \\ \theta &= 180^\circ - 135^\circ \\ \theta &= 45^\circ \end{aligned}$$

Having found θ , we can now go on to find x using sine rule which is;

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

where sides a, b, c are opposite to the angles A, B and C respectively. Knowing this and extrapolating from figure 6, we can say that;

$$\begin{aligned} \frac{x}{\sin 60^\circ} &= \frac{\sqrt{294}}{\sin 45^\circ} \\ x &= \frac{\sin 60^\circ \cdot \sqrt{294}}{\sin 45^\circ} \end{aligned}$$

Recall that;

$$\begin{aligned} \sin 60^\circ &= \frac{\sqrt{3}}{2}, \\ \sin 45^\circ &= \frac{1}{\sqrt{2}} \end{aligned}$$

¹⁷I haven't been adding my solutions for problems that involve diagrams because i have been resistant to the idea of using the images directly from mastodon and have preferred to draw th diagrams using typst but i haven't been able to do that yet so i'll be settling for this version for now until i figure out how to make geometrical diagrams like this and then i'll make a new commit.

$$\begin{aligned}
\therefore x &= \frac{\frac{\sqrt{3}}{2} \cdot \sqrt{294}}{\frac{1}{\sqrt{2}}} \\
x &= \left(\frac{\sqrt{3}}{2} \cdot \sqrt{294} \right) \div \left(\frac{1}{\sqrt{2}} \right) \\
x &= \left(\frac{\sqrt{3}}{2} \cdot \sqrt{294} \right) \cdot \sqrt{2} \\
x &= \frac{\sqrt{3} \cdot \sqrt{294} \cdot \sqrt{2}}{2} \\
x &= \frac{\sqrt{1764}}{2} \\
x &= \frac{42}{2} \\
x &= 21
\end{aligned}$$

Day 22

Find the minimum number of balls chosen from 2 red, 3 yellow, 4 green, 5 blue, 6 indigo, and 7 purple balls so that there must be five of the same color.

Solution

The first thing to do here can be to add all of the balls that are available to get the total amount of the balls.

$$\begin{aligned}
\text{total} &= 2 + 3 + 4 + 5 + 6 + 7 \\
\text{total} &= 27
\end{aligned}$$

There are 27 balls with 5, 6, 7 of them being blue, indigo and purple balls respectively. So the aim here is to find the minimum number of balls that can be picked with the condition that there must be five balls of the same color.

Let's choose blue¹⁸ to be the color that five of the balls must have. Taking this into consideration, we can remove 2 of the indigo balls leaving just 4 as well as 3 purple balls leaving 4 balls. 2 indigo balls and 3 purple balls gives us a total of 5 balls removed from 27 balls which should then leave behind 22 balls with five of these balls being of the same color and in this case, blue.

$\therefore$ The minimum number of balls chosen from 2 red, 3 yellow, 4 green, 5 blue, 6 indigo, and 7 purple balls so that there must be five of the same color is 22.

Day 24

When two six-sided dice are rolled, the probability that the sum is not divisible by 3 is $\frac{x}{36}$.

Solution

A similar problem was solved on the 15th of May. We can use the table from that problem's solution and just circle (read *bolden*) the numbers that aren't divisible by 3.

¹⁸This was arbitrarily picked. Either of indigo and purple can be picked and you'll arrive at the same answer.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Table 6: Table showing the sum of the results of two rolls of a six-sided die.

Counting the ***bolded*** numbers in the table which are obviously the numbers that aren't divisible by 3, we get a count of 24.

$\therefore$ Two six-sided die rolls sum to a number that isn't divisible by 3 with a probability of $\frac{24}{36}$ which can also be reduced to $\frac{2}{3}$.

Day 25

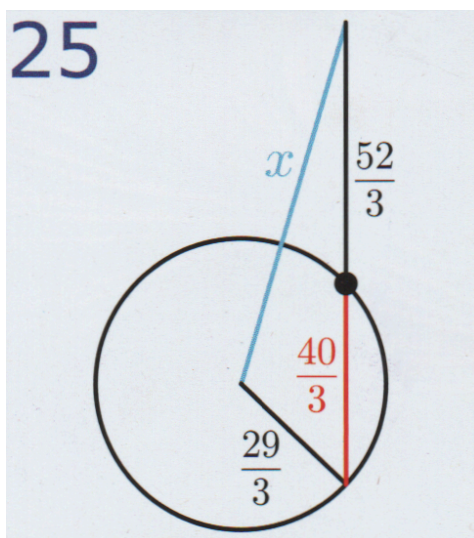


Figure 7: Problem of the day for June 25

Solution

This is a tricky problem. lol.

$$\text{let } r = \frac{29}{3}, y = \frac{40}{3}, z = \frac{52}{3}$$

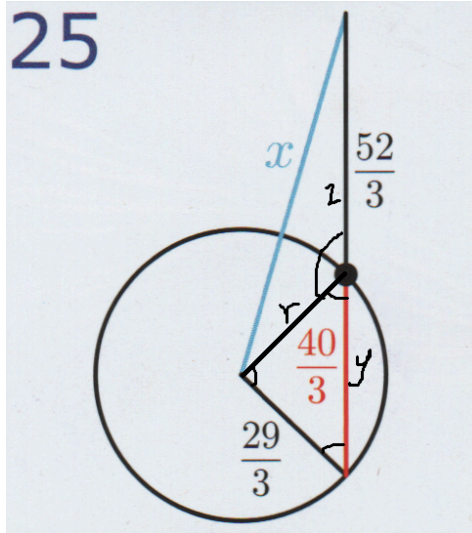


Figure 8:

From figure 8, we can find the perimeter of half the circle which is information that we can then use to find the length of the arc from the point where z and y meet to the other end of y , let's call it l which we can then use to find the angle that the arc subtends.

The perimeter of a circle, $p = 2\pi r$,

$\therefore$ the perimeter of a half circle is, $p = \pi r$,

We can then use the following ratio to find l ;

$$\frac{p}{d} = \frac{l}{y},$$

where p is the perimeter of half a circle,

and d is the diameter of the circle which is $r \times 2 = \left(\frac{29}{3}\right) \times 2 = \frac{58}{3}$

$$\frac{\pi r}{\frac{58}{3}} = \frac{l}{\frac{29}{3}}$$

Day 26

$$\frac{23}{4} \times \frac{16}{5} + \frac{7}{4} \times \frac{23}{5} - \frac{3}{4} \times \frac{3}{5}$$

Solution

Using BODMAS, we know to carry out the multiplication first before either adding or subtracting;

$$\begin{aligned}
& \frac{23}{4} \times \frac{16^4}{5} + \frac{7}{4} \times \frac{23}{5} - \frac{3}{4} \times \frac{3}{5} \\
&= \frac{368}{20} + \frac{161}{20} - \frac{9}{20} \\
&= \frac{368 + 161 - 9}{20} \\
&= \frac{529 - 9}{20} \\
&= \frac{520}{20} \\
&= 26
\end{aligned}$$

Day 27

Find the larger solution to

$$x^{\log_3 x} = 729x$$

Solution

Recall that;

$$\begin{aligned}
& \text{If } c = a^b \\
& \therefore \log_a c = b
\end{aligned}$$

We can rearrange the original equation;

$$\begin{aligned}
729x &= x^{\log_3 x} \\
\log_x 729x &= \log_3 x \\
\log_x 729 + \log_x x &= \log_3 x \\
\log_x 729 + 1 &= \log_3 x
\end{aligned}$$

We can change the log on the RHS from base 3 to base x . Recall that;

$$\begin{aligned}
\log_x y &= \frac{\log y}{\log x} \\
\therefore \log_x 729 + 1 &= \frac{\log_x x}{\log_x 3} \\
\log_x 729 + 1 &= \frac{1}{\log_x 3}
\end{aligned}$$

Multiply through by $\log_x 3$;

$$\log_x 729 \cdot \log_x 3 + \log_x 3 = 1$$

$$\log_x 3^6 \cdot \log_x 3 + \log_x 3 = 1$$

$$6 \log_x 3 \cdot \log_x 3 + \log_x 3 = 1$$

$$\text{let } a = \log_x 3$$

$$\therefore 6a \cdot a + a = 1$$

$$6a^2 + a = 1$$

$$6a^2 + a - 1 = 0$$

Solving the quadratic equation:

$$6a^2 + 3a - 2a - 1 = 0$$

$$3a(2a + 1) - 1(2a + 1) = 0$$

$$(3a - 1)(2a + 1) = 0$$

$$\therefore a = \frac{1}{3}, -\frac{1}{2}.$$

Recall that $a = \log_x 3$,

which can then be rewritten as;

$$x^a = 3$$

substituting the values of a into the equation

$$\therefore x^{\frac{1}{3}} = 3, x^{-\frac{1}{2}} = 3$$

$$x = 3^3, 3^{-2}$$

$$x = 27, \frac{1}{9}.$$

$\therefore$ The larger solution to the equation is 27.

Day 28

$$\log_2(81) \log_3(128)$$

Solution

81 and 128 can be written as follows;

$$81 = 3^4$$

$$128 = 2^7$$

$$\therefore \log_2 3^4 \log_3 2^7$$

$$= 4 \log_2 3 \cdot 7 \log_3 2$$

Recall that;

$$\log_x y = \frac{\log y}{\log x}$$

$$\begin{aligned}
&= 4 \log_2 3 \cdot 7 \left\lceil \frac{\log_2 2}{\log_2 3} \right\rceil \\
&= 4 \log_2 3 \cdot 7 \left\lceil \frac{1}{\log_2 3} \right\rceil \\
&= \cancel{4 \log_2 3} \cdot 7 \left\lceil \frac{1}{\cancel{(\log_2 3)}} \right\rceil \\
&= 4 \times 7 \\
&= 28
\end{aligned}$$

Day 29

$$\begin{aligned}
y - \frac{1}{y} &= 5 \\
x &= \left(y + \frac{1}{y} \right)^2
\end{aligned}$$

Solution

We can start by squaring both sides of the equation;

$$\left(y - \frac{1}{y}\right)^2 = 5^2$$

$$y^2 - \frac{y}{y} - \frac{y}{y} + \frac{1}{y^2} = 25$$

$$y^2 - 1 - 1 + \frac{1}{y^2} = 25$$

$$y^2 - 2 + \frac{1}{y^2} = 25$$

$$y^2 + \frac{1}{y^2} = 25 + 2$$

$$y^2 + \frac{1}{y^2} = 27$$

Now, for x ;

$$x = \left(y + \frac{1}{y}\right)^2$$

$$x = \left(y + \frac{1}{y}\right)\left(y + \frac{1}{y}\right)$$

$$x = y^2 + \frac{y}{y} + \frac{y}{y} + \frac{1}{y^2}$$

$$x = y^2 + 1 + 1 + \frac{1}{y^2}$$

$$x = y^2 + 2 + \frac{1}{y^2}$$

$$y^2 + \frac{1}{y^2} = x - 2$$

Equating both values for $y^2 + \frac{1}{y^2}$

$$x - 2 = 27$$

$$x = 27 + 2$$

$$x = 29$$

July

Day 4

How many of the number from 2 to 50 have exactly 3 factors?

Solution

I am not the most creative mathematician so naturally, i can't think of a creative way for solving this problem. The only way to do this that comes to mind is to go through each number from 2 to 50 and check for the factors of each number and then count the numbers that have exactly 3 factors. This is a lot of work to do by hand so why not write a program that handles this for us.

```
//that is right. i have to import ArrayList.
import java.util.*;

public class July4 {

    public static void main(String[] args) {
        System.out.println(
            "Setting up the program. all systems are a go for launch."
        );

        int x = 2;
        int l = 0; //a variable to keep track of the numbers that have three factors.
        ArrayList<Integer> arr = new ArrayList<>(); // a list to add all of the numbers that meet our
        condition.
        while (x != 51) {
            int k = 0; //this variable is for keeping track of the ammount of times x is divisble by a
            number i.e. a factor.

            //a for loop to check if there is any remainder when you divide x by all of the numbers from 1
            to x.
            for (int j = 1; j <= x; j++) {
                if (x % j == 0) {
                    k++;
                }
            }

            //if k(the factor(s)) is equal to three, then you increment l by one and add x which is the
            number that meets this condition to the list.
            if (k == 3) {
                l++;
                arr.add(x);
            }

            //increment x by one to keep the while loop going.
            x++;
        }

        System.out.println(
            "The amount of numbers from 2 to 50 that have exactly 3 factors is " +
            l
        );
        System.out.println(
            "the set of numbers from 2 to 50 that have exactly 3 factors is " +
            arr
        );
    }
}
```

When you run the program,¹⁹ you get the following output in your terminal;

Setting up the program. all systems are a go for launch.

The amount of numbers from 2 to 50 that have exactly 3 factors is 4

the set of numbers from 2 to 50 that have exactly 3 factors is [4, 9, 25, 49]

I went an extra step with my program. Instead of just finding the amount of numbers that meet the condition, we could get the numbers that do meet this condition and which—as we can see from terminal output—are 4, 9, 25, 49.

Coincidentally, the numbers happen to be perfect squares whose square roots are 2, 3, 5, 7. If you look further at this numbers, you can see that they are the first four prime numbers.

∴ The amount of numbers from 2 to 50 that have exactly 3 factors is 4

Day 5

$$\frac{\sqrt{63} + \sqrt{28}}{\sqrt{63} - \sqrt{28}}$$

Solution

This is pretty straightforward. We use the factors of these numbers to reduce the equation;

$$\begin{aligned} &= \frac{\sqrt{7 \times 9} + \sqrt{7 \times 4}}{\sqrt{7 \times 9} - \sqrt{7 \times 4}} \\ &= \frac{\sqrt{7} \times \sqrt{9} + \sqrt{7} \times \sqrt{4}}{\sqrt{7} \times \sqrt{9} - \sqrt{7} \times \sqrt{4}} \\ &= \frac{3\sqrt{7} + 2\sqrt{7}}{3\sqrt{7} - 2\sqrt{7}} \\ &= \frac{5\cancel{\sqrt{7}}}{\cancel{\sqrt{7}}} \\ &= 5 \end{aligned}$$

¹⁹I first wrote the pseudocode in my notebook to flesh out the logic before jumping to my IDE²⁰ to write the code. My code was sound but i kept on having issues with getting the final answer which is 4. It turned out that my understanding of factors was incomplete so i went to the encyclopedia of mathematics because my internet was giving me issues at the time and i updated my understanding of factors with the entry on factors. I thought the factors of a number didn't include 1 and the number itself.

According to the encyclopedia of mathematics, the factor(s) of a number, x , is a number, y , that x is divisible by without any remainders including both 1 and x . I don't know where, along the line in my mathematical journey, I made this assumption of 1 and x not being factors and what might have caused me to make such an assumption.

²⁰which is Zed btw.

The order in which these problems were solved.

- Day 14.
- Day 9.
- Day 17.
- Day 16.
- Day 7.
- Day 20.
- Day 22.
- Day 23.
- Day 24.
- Day 26.
- Day 31.
- Day 2.
- Day 14.
- Day 15.
- Day 16.
- Day 20.
- Day 22.
- Day 25.
- Day 26.
- Day 28.
- Day 30.
- Day 5.
- Day 14.
- Day 15.
- Day 20.
- Day 26.
- Day 8.
- Day 4.
- Day 29(May).
- Day 17.
- Day 24.
- Day 22.
- Day 20.
- Day 21.
- Day 26.
- Day 27.
- Day 25.
- Day 29.
- Day 28.
- Day 15.
- Day 10.
- Day 4(July).
- Day 5.